

QuoteFlow AI - Detailed Project Specification

QuoteFlow AI - Master Project Document

Comprehensive specification for an AI-powered quotation, billing, CRM, receipt, analytics and marketing automation platform for Indian businesses.

1. Vision

Build a unified platform that replaces separate billing software, CRM tools, quotation generators, marketing tools and reporting systems.

2. Business Objectives

Reduce quotation creation time by 90%, automate billing workflows, increase lead conversion, improve payment collection and provide AI-driven business insights.

3. Problem Statement

Businesses struggle with manual quotation creation, inconsistent branding, scattered customer records, poor follow-up management and lack of actionable analytics.

4. Target Users

Software companies, agencies, freelancers, wholesalers, retailers, manufacturers, consultants and startups.

5. Core Modules

Authentication, Company Management, Customer Management, Product Catalog, Quotation Management, Invoice Management, Receipt Management, CRM, Lead Tracking, Marketing Automation, Analytics, AI Assistant, Notifications and Subscription Management.

6. Detailed Tech Stack

Frontend: Flutter. Backend: Spring Boot Java 21. Database: PostgreSQL. Cache: Redis. Storage: AWS S3. Queue: Kafka. Authentication: JWT + OAuth. Notifications: Firebase. Monitoring: Grafana and Prometheus. CI/CD: GitHub Actions.

7. System Architecture

Mobile App and Web App communicate with API Gateway. Gateway routes requests to Auth, CRM, Billing, Quotation, Marketing and AI services. Services use PostgreSQL, Redis and S3.

8. Database Design

Entities include Users, Roles, Companies, Customers, Products, Quotations, Quotation Items, Invoices, Invoice Items, Receipts, Leads, Activities, Campaigns, Payments and Audit Logs.

9. Functional Requirements

Users can create quotations, generate PDFs, convert quotations to invoices, record payments, generate receipts, manage leads, schedule campaigns and analyze business performance.

10. Non Functional Requirements

Response time below 2 seconds, 99.9% uptime, encrypted data storage, role-based authorization and scalable architecture.

11. Quotation Flow

Create customer, add products/services, apply GST, generate quotation number, create PDF, send through WhatsApp, email or share sheet.

12. Invoice Flow

Approved quotation converts into invoice. Invoice receives unique number, payment status tracking and receipt generation.

13. Marketing Flow

Select audience, choose template, schedule campaign, send through WhatsApp, SMS or email and track delivery metrics.

14. AI Features

AI quotation generator, proposal writer, pricing assistant, lead scoring, sales forecasting, customer sentiment analysis and business recommendations.

15. Security Design

JWT authentication, AES-256 encryption, HTTPS, audit logs, IP monitoring, brute force protection, password hashing using BCrypt.

16. Edge Cases

Duplicate GST, duplicate invoice numbers, network loss, failed payment gateway, PDF generation failure, storage limits, expired sessions and invalid customer records.

17. Error Handling

Centralized exception management, user-friendly messages, retry mechanisms, logging and alerts.

18. API Design

REST APIs with versioning, pagination, filtering, validation, rate limiting and OpenAPI documentation.

19. Deployment

Development, Staging and Production environments using Docker, Kubernetes and AWS infrastructure.

20. Testing Strategy

Unit testing, integration testing, API testing, UI testing, performance testing, security testing and UAT.

Detailed Test Cases

TC-AUTH-001

Scenario: Login with valid OTP

Expected Result: User successfully logged in

TC-AUTH-002

Scenario: Login with invalid OTP

Expected Result: Validation error shown

TC-COMP-001

Scenario: Create company

Expected Result: Company saved

TC-CUST-001

Scenario: Create customer

Expected Result: Customer saved

TC-CUST-002

Scenario: Duplicate GST

Expected Result: System blocks duplicate

TC-QUOTE-001

Scenario: Create quotation

Expected Result: Quotation generated

TC-QUOTE-002

Scenario: Generate PDF

Expected Result: PDF downloaded

TC-QUOTE-003

Scenario: Share quotation

Expected Result: Share sheet opened

TC-INV-001

Scenario: Convert quotation to invoice

Expected Result: Invoice generated

TC-INV-002

Scenario: Apply GST

Expected Result: Correct tax calculated

TC-PAY-001

Scenario: Record payment

Expected Result: Payment stored

TC-REC-001

Scenario: Generate receipt

Expected Result: Receipt PDF created

TC-MKT-001

Scenario: Send campaign

Expected Result: Campaign delivered

TC-MKT-002

Scenario: Schedule campaign

Expected Result: Campaign scheduled

TC-ANA-001

Scenario: Revenue report

Expected Result: Report generated

TC-SEC-001

Scenario: SQL Injection

Expected Result: Blocked

TC-SEC-002

Scenario: JWT Tampering

Expected Result: Unauthorized

TC-PERF-001

Scenario: 1000 concurrent users

Expected Result: System remains stable

TC-UAT-001

Scenario: End-to-end quotation workflow

Expected Result: Success

TC-UAT-002

Scenario: End-to-end billing workflow

Expected Result: Success